Is the growth sustainable, or just expensive?

A diagnostic of growth quality across channel unit economics, cohort activation and retention, randomized incrementality, observed price response, segment and product profitability, customer concentration, and a bounded spend-reallocation scenario.

Prepared by Miguel Fidalgo Martins · Revenue Analytics

Coverage 36 months, January 2023 to December 2025 · 9,000 customers · 69,950 transactions · $5.6M acquisition spend

Data Deterministic synthetic case study; figures are not a real-company forecast

Contents

1 Executive summary

Case conclusion. Paid search and social ads hold 68% of the acquisition budget while both return less than 1× CAC. Under the stated response assumptions, a budget-neutral reallocation increases modeled observed-window contribution by $7.9M (48%). The result remains positive in the three defined stress cases.

Average monthly revenue in the final six months is 9.6× the first six-month average. Customer volume explains most of the change, while the acquisition mix concentrates spend in the two weakest channels. Section 5.8 models a bounded reallocation; it is a decision scenario, not a forecast.

$54.6M	$16.6M	20.3×	0.43×
Total revenue	**Contribution margin**	**Best channel LTV/CAC**	**Worst channel LTV/CAC**
36-month window	30.3% of revenue	organic · 0.0m payback	social ads · loses money

Monthly revenue rose from $65K in January 2023 to $3.1M in December 2025, and contribution margin tracked it closely in absolute dollars. The channel, cohort, decomposition, and margin views qualify that top-line growth: acquisition allocation and early customer activity remain the material constraints in this synthetic case.

First, channel economics split sharply with no middle ground. Organic, referral, and partners return between 4.4 and 20.3 times their acquisition cost and recover CAC in the mature-cohort curve within two months. Paid search and social ads return 0.93 and 0.43 times cost, so the average customer they buy is worth less than the cost to acquire them. These two channels hold $3.78M, or 68%, of total acquisition spend, while generating only 16% of channel contribution. The three efficient channels do the inverse: 25% of spend, 78% of contribution.

Second, growth is volume-led rather than monetization-led. Of the $15.1M revenue increase between the first and last six months, 69% comes from acquiring more customers and only 25% from higher revenue per customer. Volume bought through channels that lose money is the weakest form of growth, because holding the line depends on continued spend.

Third, activation and subsequent activity are weak. Median month-0 activation is 52% of signups. Median signup activity is 34% at month 6 and 22% at month 12. Among customers active in month 0, 36% remain active at month 6. Median cohort revenue retention is 68% at month 6 and 50% at month 12. These cohort-level diagnostics do not identify a causal mechanism.

Fourth, margin concentrates in the least efficient places. Enterprise is the largest segment at 48% of revenue but carries the lowest margin rate at 26.2%, the only segment below the 30% quality floor. The Services product line runs at just 17.7% margin on $13.6M of revenue, and Premium also sits below the floor at 24.9%. Regional margins, by contrast, are banded within 1.0 points, so geography is not the problem. Mix and cost-to-serve are.

Fifth, revenue is highly concentrated. The top 10% of customers account for 64% of revenue and the top 20% for 81%, with a Gini coefficient of 0.77. That concentration makes retention of high-value customers a first-order risk, and it makes the early cohort decay more expensive than the median retention figure suggests.

The modeled response is a bounded reallocation, governed as a test-and-scale motion. Scaling the three efficient channels within guardrails and pulling 35% from the two inefficient ones lifts observed-window contribution from $16.6M to $24.5M, an uplift of $7.9M (48%) at the same acquisition budget. It is positive in the defined stress cases, from $2.2M worst to $12.8M best, and across 5 deterministic seeds it averages $8.0M (coefficient of variation 1.3%). Full recommendations are in Section 7, rollout controls in Section 8.

2 Context and objectives

Top-line revenue growth is the metric most often celebrated and least often interrogated. A revenue line that bends upward can hide three separate problems at once: customers acquired below the cost to serve them, cohorts that leak revenue faster than new cohorts replace it, and margin diluted by an unfavorable product or segment mix. None of these appear in a revenue chart. All three change what a business should do next.

This analysis exists to answer budget and pricing questions, not to populate a dashboard. The business is spending $5.6M across the observed period to acquire customers and wants to know whether that spend is building a durable asset or renting a revenue line that stops the moment spend stops. The question is asked through complementary descriptive, causal, and scenario methods.

Question	Method	Decision it informs
Is growth converting into margin?	Monthly revenue and contribution margin trend, margin rate vs floor	Whether to defend or improve margin
What is driving the growth?	Decomposition into volume, monetization, and mix	Where to invest for durable growth
Do cohorts activate and hold value?	Month-0 activation, signup activity, retained month-0 customers, and revenue retention	Whether activation or retention needs diagnosis
Which channels deserve budget?	LTV/CAC, payback, and spend-to-contribution alignment per channel	How to allocate acquisition spend
Which marketing activity is incremental?	Randomized customer holdouts with CUPED-adjusted contribution lift	Which treatments merit a scaled experiment
How does demand respond to price?	Randomized weekly price assignments with fixed effects and week-clustered uncertainty	Which bounded price candidates to test
How should observed value be allocated across touches?	Fully reconciling position-based attribution	Descriptive journey allocation, not causal lift
What is the upside from reallocating spend?	Bounded scenario with stress and seed-stability tests	Size and confidence of the reallocation

The scope is deliberately narrow. This is not a forecast or market sizing. It is a diagnostic of growth quality with explicit causal and non-causal claim boundaries, built so that each finding either confirms the current allocation or changes it. Where a finding cannot be settled with the data on hand, Section 6 says so plainly rather than papering over the gap.

The decision standard is practical: a finding is strong enough to change budget only when it is visible in at least two places, such as channel economics and spend mix, or cohort decay and customer concentration. Findings supported by one view only are treated as monitoring items or experiments, not as immediate allocation moves. That is why the report recommends a controlled reallocation, a retention workstream, and a margin diagnostic rather than a single blanket growth call.

Who should read this

The primary audience is an executive or commercial owner reviewing acquisition allocation. The secondary audience is the analytics or finance reviewer who needs to trust the numbers before acting on them. The report is written for the first audience in the prose and for the second in the methodology and appendix.

3 Data and methodology

The pipeline validates six raw contracts before building the analytical outputs. The commercial base has 9,000 customers, 69,950 transactions, and 6,576 daily channel-spend records from January 2023 to December 2025. The measurement layer adds 22,465 privacy-minimal touchpoints, 3,000 randomized marketing observations, and 1,680 randomized weekly pricing cells. dbt builds tested incremental facts and marts; pandas produces empirical payback, completed cohort grids, causal estimates, scenarios, and publications. Final QA reconciles the warehouse, Python, dashboard, and report outputs.

Metric definitions

Contribution margin is revenue minus direct delivery cost. Observed lifetime value is the mean cumulative contribution margin per acquired customer, computed across the full base including the 843 customers (9.4%) who never transacted, so the figure is not inflated by survivorship. Customer acquisition cost is period-level channel spend divided by the customers acquired in that channel. The LTV-to-CAC ratio divides the two. Payback uses customers mature enough to reach 24 acquisition-age months, includes mature zero-transaction customers, and records the first month when cumulative contribution per mature customer recovers a time-aligned CAC: spend between the first and last signup dates of that mature subset divided by its customers. A channel that does not recover CAC is reported as >24 months rather than assigned an invented point estimate.

A channel is classified efficient when LTV/CAC is at least 3.0 and payback is 12 months or less. It is inefficient when LTV/CAC is below 1.0, observed payback exceeds 24 months, or CAC is not recovered inside the horizon. Insufficient maturity remains undefined. The margin quality floor is 30%. These thresholds live in one metric registry imported by the analysis, API, chart pack, and validation gate. Cross-output checks fail when a consumer diverges from the registry.

Causal and attribution methods

Marketing incrementality uses randomized customer holdouts and CUPED adjustment from pre-period contribution. Treatment-minus-control lift is reported per treated customer with a 95% confidence interval. The tested channels are Paid Search, Social Ads; the combined interval bounds across the two experiments run from $1.54 to $41.11 per treated customer.

Price elasticity is estimated from randomized product-region-week assignments at price indices 0.90, 1.00, and 1.10. Log demand is regressed on log price with region and week-of-year fixed effects and CR1 uncertainty clustered by intervention week. Product coefficients range from -1.58 to -0.59. Pricing candidates remain inside the observed range. Position-based multi-touch attribution separately allocates $16,564,031 of observed contribution and reconciles to the customer table; it is descriptive and is not used as an incrementality estimate.

The report uses average and median LTV together because they answer different questions. Average LTV is the right numerator for contribution economics because it reconciles to total contribution. Median LTV is the buyer-quality check because it shows the typical customer experience underneath the mean. When both measures point in the same direction, the channel conclusion is stronger; when they diverge, the channel needs segmentation before scaling.

Cohort construction

Customers are grouped into monthly signup cohorts. Month-0 activation is active customers divided by all signups. Signup activity at age m uses that same signup denominator. Retained-from-month-0 activity instead asks what share of month-0 active customers are also active at age m. Revenue retention divides cohort revenue at age m by cohort revenue in month 0. Reported curves use the median across cohorts at each age; later ages include only cohorts old enough to be observed.

Revenue decomposition

The growth decomposition compares the first six months of the window against the last six. The total revenue change is split into a customer-volume effect, an average-revenue (monetization) effect, and a segment-mix effect, with any unexplained difference carried as a residual. The residual is zero here by construction, so the three terms reconcile the arithmetic change under this specification; they do not exhaust its causal mechanisms. Segment is the mix dimension; other definitions of mix, such as region or product, would allocate the mix term differently, which Section 6 notes.

Scenario engine

The reallocation scenario is deliberately conservative. It applies bounded CAC and LTV elasticities to each channel, caps any channel's scale-up at 100% of current spend, and holds back budget rather than forcing it into channels that have run out of efficient headroom. The scenario is stress-tested across best, base, and worst elasticity cases, and repeated across 5 deterministic seeds to measure sensitivity to draws from the same generator. This does not establish external validity. The acquisition budget is held constant in every case; other operating costs and capacity constraints are outside the scenario.

The scenario is not allowed to create value by expanding the budget, deleting weak channels, or assuming unlimited capacity in strong ones. It must earn the uplift through reweighting the existing budget and absorbing weaker economics as spend scales. The base case is a modeled planning case and the worst case is one specified downside case, not a guaranteed floor.

Data quality gate

All 15 raw-data validation checks pass before any analysis runs: schema match, grain uniqueness, referential integrity, date coverage, channel-domain alignment, and value-range sanity. The gate flagged 258 transactions (0.37%) where cost exceeds revenue. That share sits below the 1% review threshold, so the intentional synthetic cost-to-serve exceptions are retained rather than filtered. At the customer level, 22 customers carry a negative lifetime contribution margin, 0.2% of the base, which the LTV figures absorb rather than exclude. The full gate is reproduced in Appendix F.

Limitations of the data

The data is synthetic. It is built to be realistic and internally consistent, which makes it suitable for demonstrating method and for relative comparison between channels, segments, and cohorts. It is not a forecast of any real market. Observed LTV reflects only the window available, so long-horizon cohort reads draw on mature cohorts only. These limits are revisited in Section 6.

4 Analytical framework

Growth quality is not a single metric. It is the answer to whether the next dollar of revenue is worth more, the same, or less than the average dollar already on the books. This report builds that answer from four layers, each one narrowing the question the previous layer leaves open.

The first layer is the **aggregate trajectory**: does revenue growth carry margin growth, and does the margin rate improve, hold, or erode as the business scales. A flat margin rate during rapid growth is the first signal that incremental revenue is no better than average, and it sets up every question that follows.

The second layer is **attribution of the growth**: whether the increase comes from more customers, more revenue per customer, or a shift in mix. Volume-led growth and monetization-led growth demand different responses. The decomposition separates them so the response can be matched to the cause.

The third layer is **unit economics and durability**: whether the customers being acquired are worth more than they cost, and whether they stay long enough to pay back. This is where channel LTV/CAC, payback, and cohort evidence combine. A low CAC is not sufficient if cumulative cohort contribution does not recover it inside the governed horizon.

The fourth layer is **concentration and structural risk**: where revenue and margin pool, and how exposed the business is to a small set of customers, segments, or products. Concentration is not inherently bad, but it changes which risks matter and which levers move the margin line.

The reallocation scenario sits on top of all four. It takes the channel economics from layer three, applies conservative elasticities, and tests how much contribution a budget-neutral move can add. The framework is built so that the scenario is a consequence of the findings, not an assertion bolted onto them.

> Read the findings as a single argument. Each section answers the question the previous one raises, and the reallocation in Section 5.8 is the action the first seven findings point to.

5

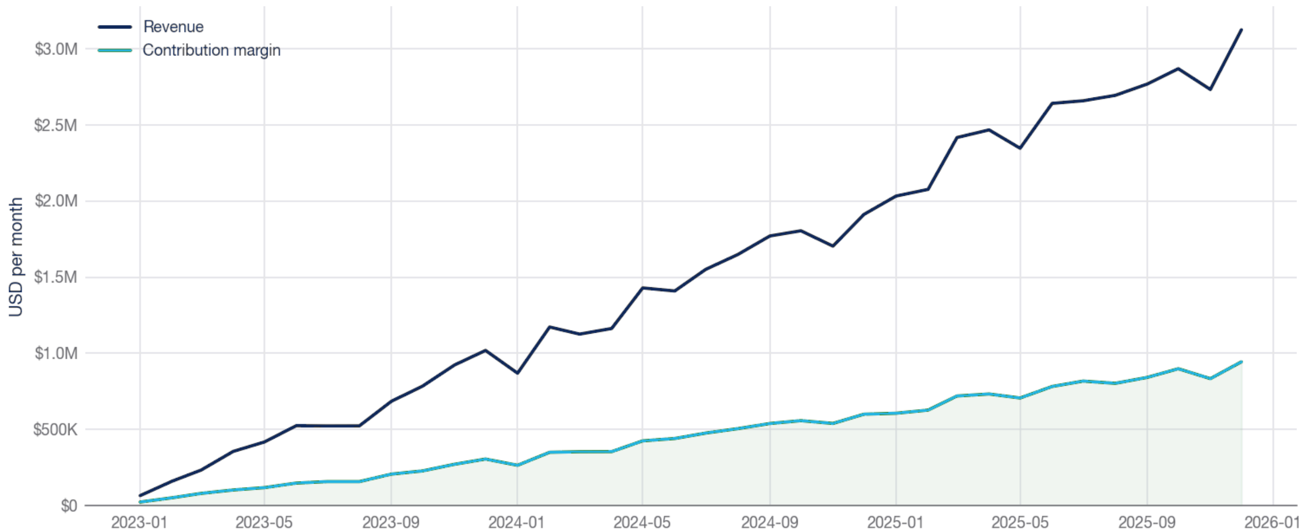
Findings

5.1 Observed growth is fast and uneven

Monthly revenue rose from $65K in January 2023 to $3.1M in December 2025. Comparing like with like, the last six months averaged $2.8M a month against $293K in the first six, a 9.6-fold increase across 36 months. Contribution margin also rises in absolute dollars, while the margin rate remains close to its starting level.

Is growth converting into margin?

Monthly revenue and contribution margin, full coverage window.



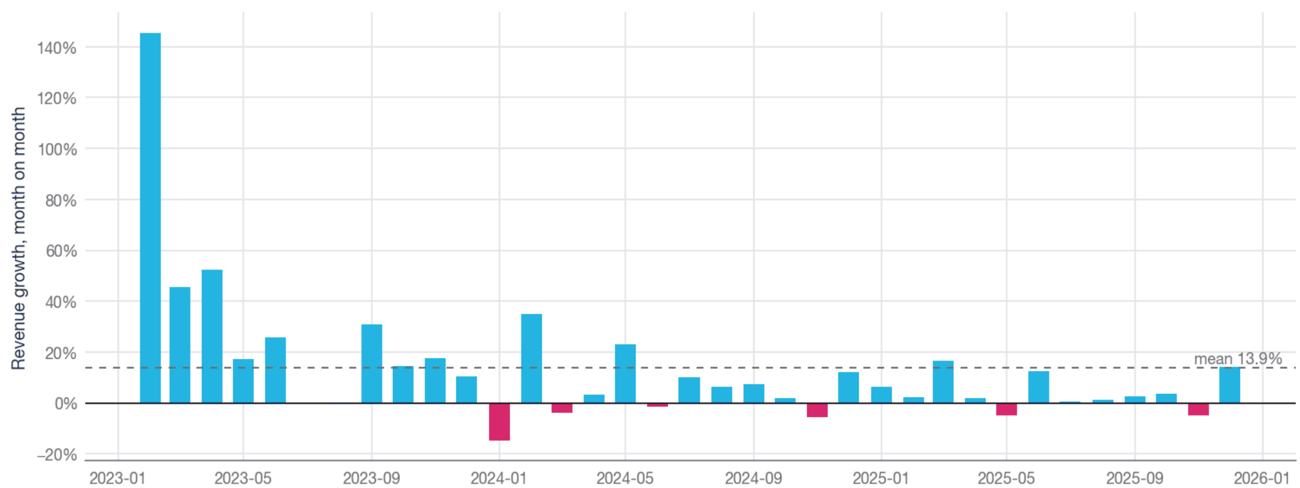
Source: monthly_revenue_health.csv · synthetic data

Figure 1. Revenue and contribution margin rise together in absolute dollars. The gap between the two lines, which is the margin rate, does not widen as the business scales.

Month-on-month revenue change averages 13.9%, with 7 down months in the observed window. The chart shows recurring dips, but three years of synthetic data are not enough to separate seasonality from other generator effects.

How steady is the growth?

Month-on-month revenue change. Down months are the test of durability.



Source: monthly_revenue_health.csv · synthetic data

Figure 2. Month-on-month revenue growth, including 7 observed down months. No causal or seasonal model is fitted.

The active base grew from 67 customers in the opening month to 2,245 in the closing month, while revenue per active customer moved from $976 to $1,392. The decomposition in Section 5.3 quantifies the relative volume and monetization effects.

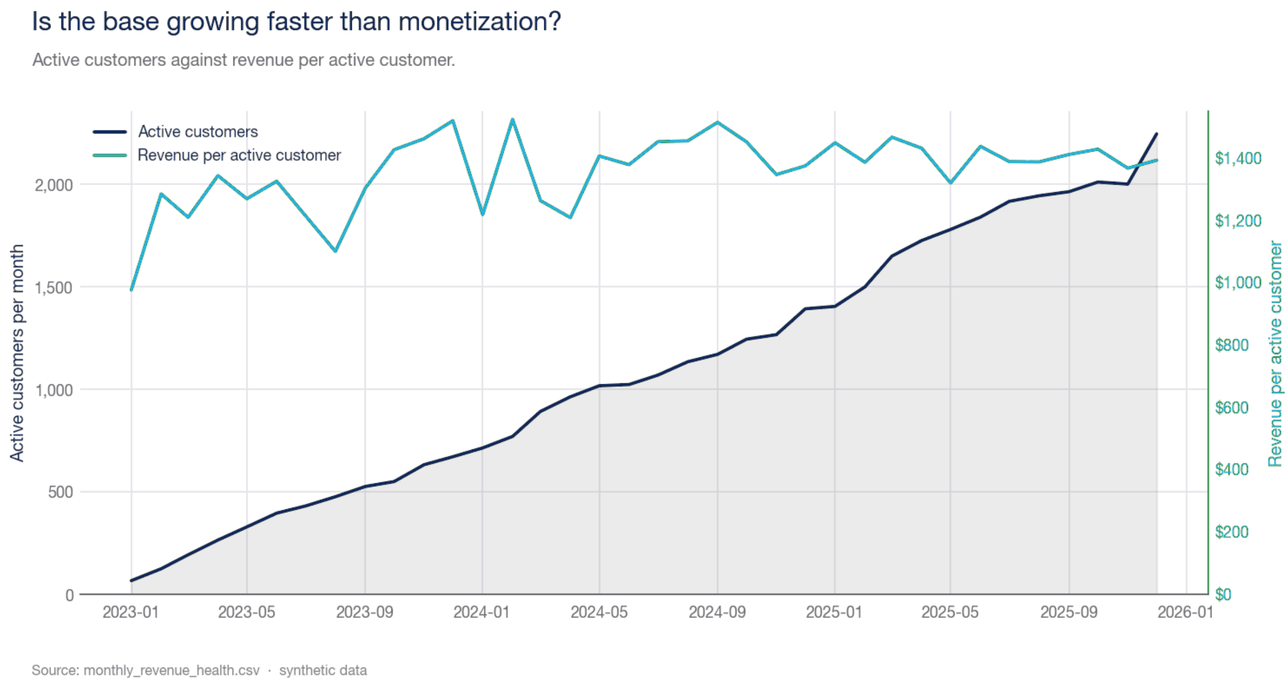


Figure 3. The active base climbs steeply while revenue per active customer stays in a narrow band. Growth is coming from more customers, not richer ones.

The active base expanded 33.5× from the first month to the last; revenue per active customer rose only 43%. Channel CAC and cohort activity therefore need to accompany the top-line measure in any allocation decision.

5.2 Margin quality is only holding, not improving

Contribution margin holds between 28% and 35%, close to the governed 30% review floor. Absolute margin grows, but the rate shows no material upward movement in the observed series.

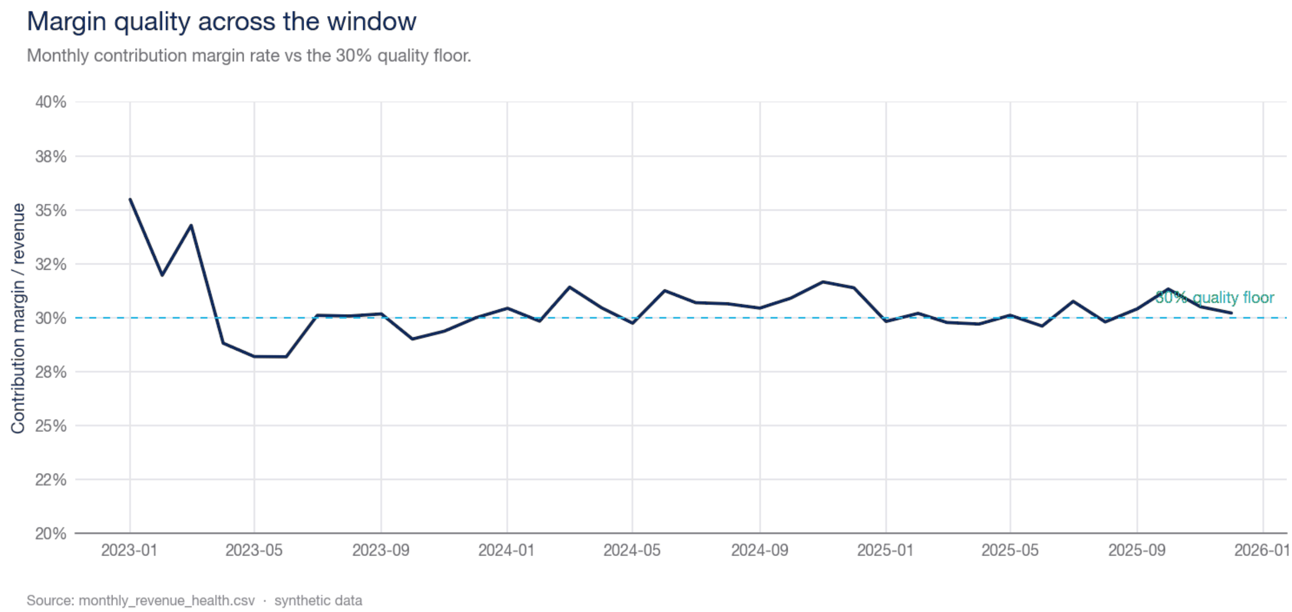


Figure 4. Monthly contribution margin rate against the 30% quality floor. The rate oscillates around the floor for the full window and shows no upward drift.

The aggregate rate alone cannot identify whether channel allocation, product mix, pricing, or cost-to-serve explains the pattern. The channel and profitability cuts below locate the largest diagnostic gaps without claiming causality.

5.3 Growth is volume-led, not monetization-led

Decomposing the $15.1M revenue change between the first and last six months separates three effects: more customers, more revenue per customer, and a shift in segment mix.

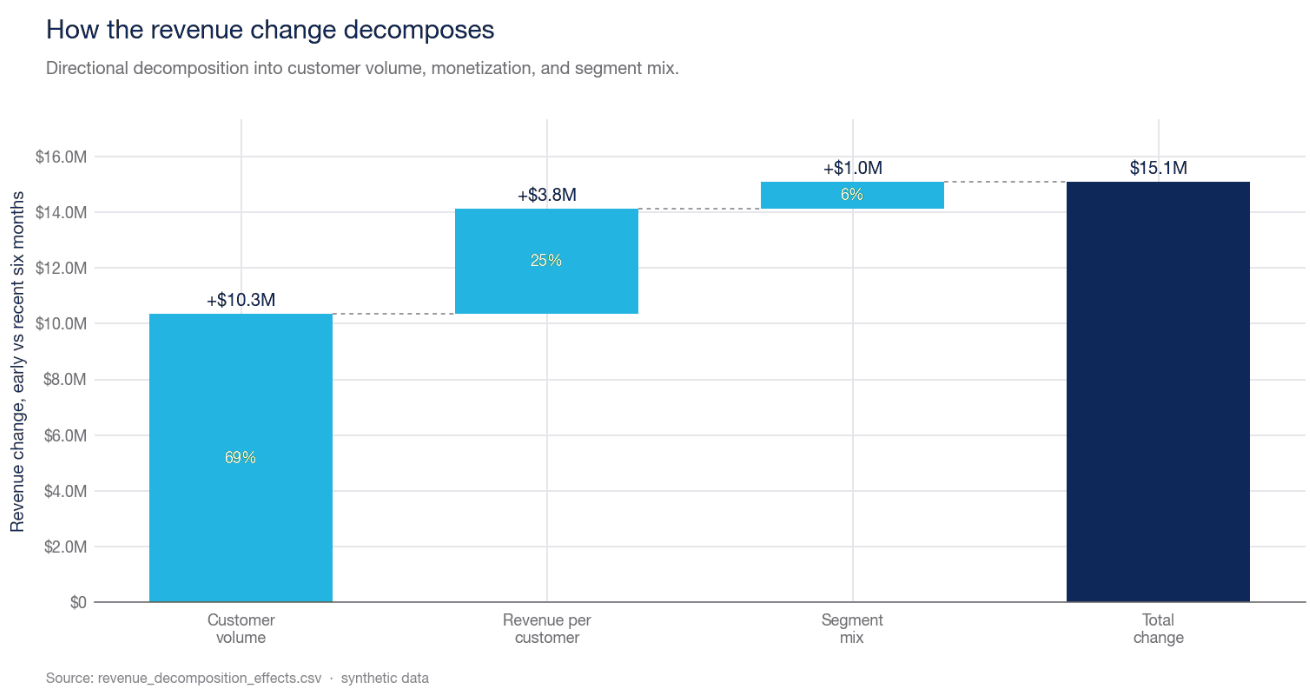


Figure 5. Volume accounts for 69% of the revenue change, monetization 25%, and segment mix 6%. The three terms reconcile the change under the chosen segment-mix decomposition.

Effect	Contribution to change	Share
Customer volume	$10,349,685	68.6%
Revenue per customer	$3,771,454	25.0%
Segment mix	$966,809	6.4%
Total change	**$15,087,948**	**100.0%**

Volume accounts for 69% of the increase. Monetization adds 25% and mix a further 6%. This decomposition does not attribute incremental customers to acquisition channels; it establishes that volume, rather than per-customer revenue, dominates the window-to-window change.

The mix effect is small at 6%, which is its own finding. The business is not growing into a richer customer mix that would lift the rate on its own. If anything, the structural margin problem documented in Section 5.6 means a larger mix shift toward the biggest segment would pull the rate down, not up.

For planning, that makes buyer quality and acquisition efficiency material guardrails. The channel view in Section 5.5 evaluates the current portfolio; it should not be read as causal attribution for the historical decomposition.

5.4 Activation and cohort activity weaken early

Median month-0 activation is 52% of signups. At month 6, 34% of all signups are active; among customers active in month 0, 36% are also active at month 6. The corresponding month-12 measures are 22% and 24%. These denominators are kept separate to avoid describing late activation as retention.

Median cohort revenue retention is 88% at month 3, 68% at month 6, and 50% at month 12. The change from month 3 to 6 is 20.1 percentage points; the change from month 6 to 12 is 18.7 points.

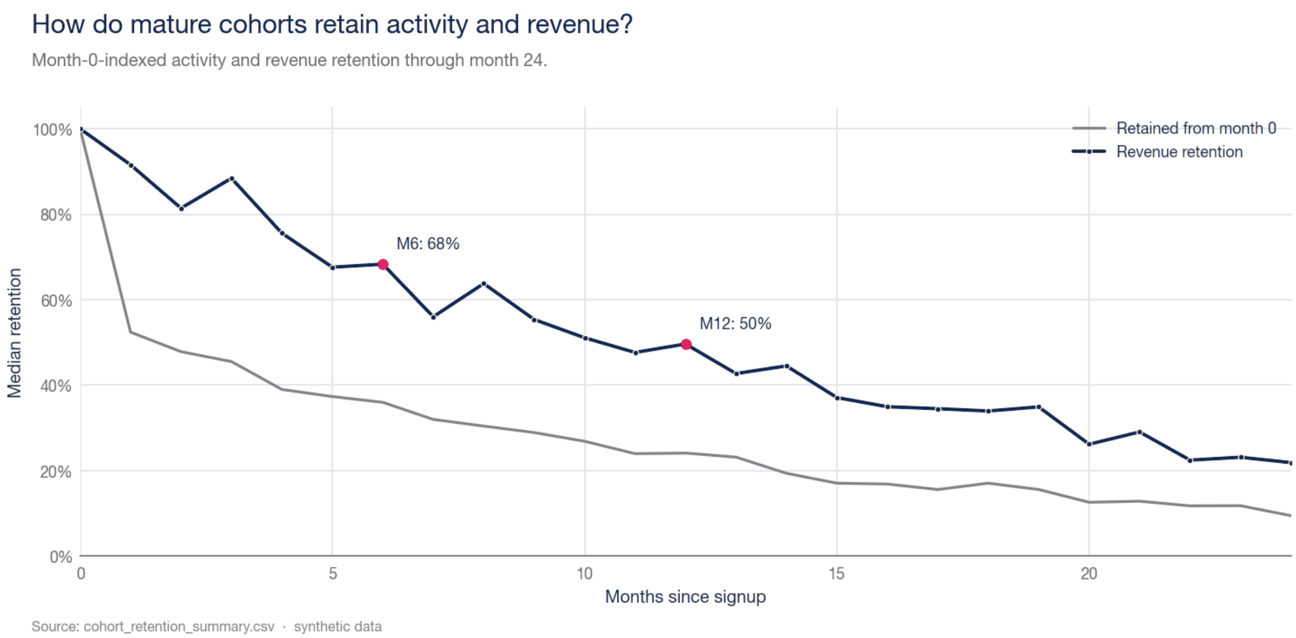


Figure 6. Median retained-from-month-0 activity and revenue retention through month 24.
Signup activity uses a different denominator and is reported in the text and appendix.

16.7% of cohorts with a month-6 observation exceed their own month-0 revenue at that age. Revenue retention above retained-customer activity can reflect higher spend among active customers, reactivation, or composition change; this aggregate table does not distinguish those mechanisms.

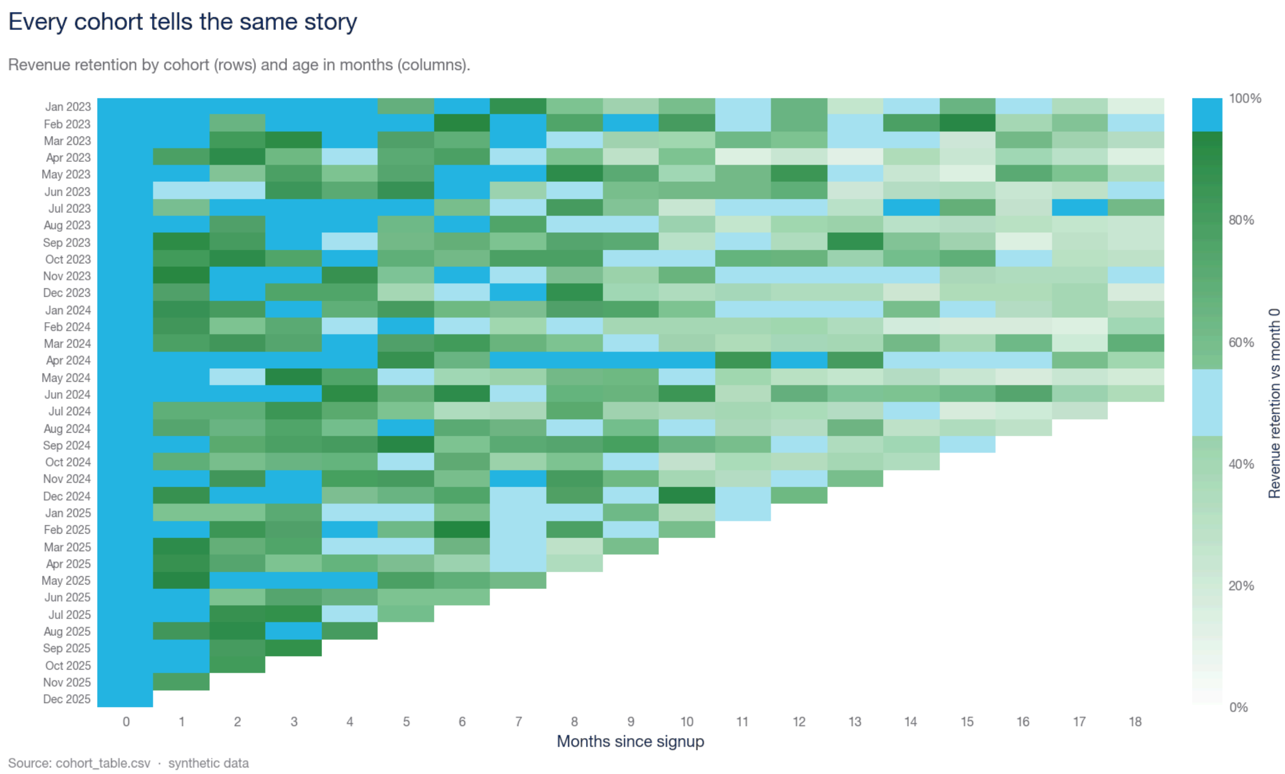


Figure 7. Revenue retention by cohort (rows) and age (columns). The heatmap shows both the recurring decay pattern and cross-cohort variation.

The evidence supports a follow-up activation and retention diagnostic by channel, segment, and product. The aggregate cohort curve is not joined to payback evidence at customer level, so it does not prove that a particular activity drop causes a channel to miss payback.

The month-6 read is the most decision-useful point on the curve. It is mature enough to include 30 cohorts and early enough to support a practical leading indicator. By contrast, the month-24 tail uses 12 mature cohorts and carries greater sampling uncertainty.

5.5 Two channels fail the unit-economics policy and hold most of the budget

Channel economics split sharply, while most acquisition spend sits in the two channels below 1× observed LTV/CAC.

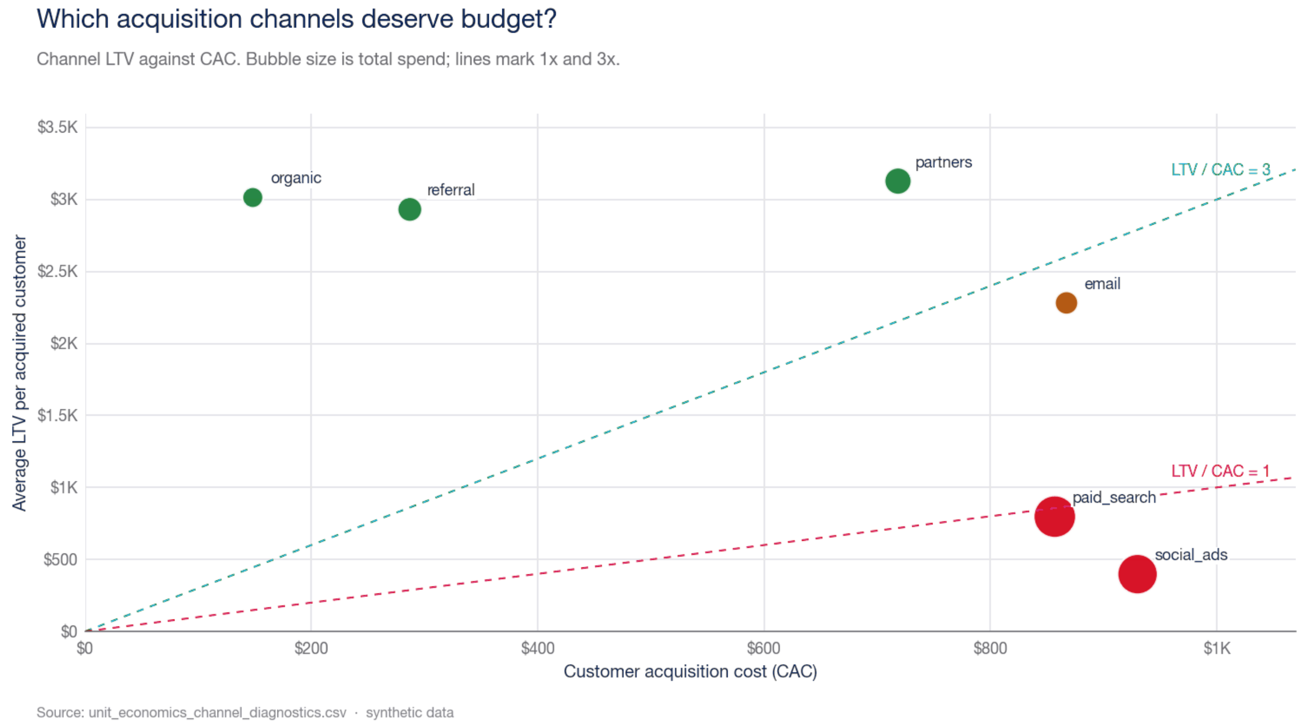


Figure 8. Channel LTV against CAC, with bubble size showing total spend. Three channels sit above the 3× line and two below 1×. The two largest bubbles, paid search and social ads, are below the CAC-recovery line on observed LTV.

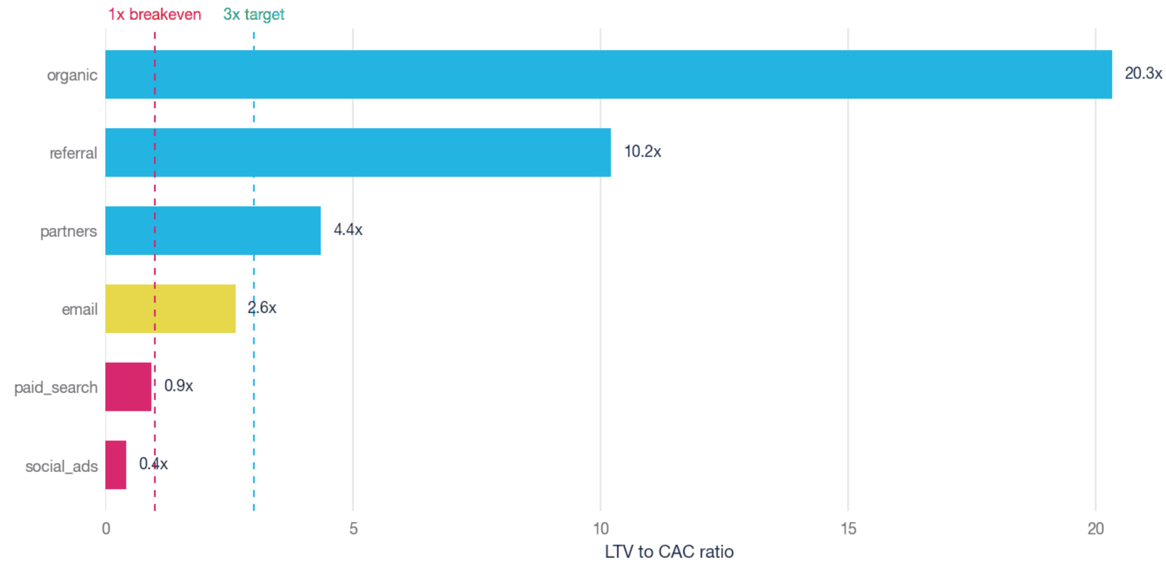
Channel	Customers	Spend	CAC	Avg LTV	Med LTV	LTV/CAC	Payback evidence (mo)	Status
organic	1,736	$257,299	$148	$3,014	$984	20.33	0.0	efficient
referral	1,668	$478,649	$287	$2,931	$1,117	10.21	1.0	efficient
partners	889	$638,558	$718	$3,127	$1,085	4.35	3.0	efficient
email	455	$394,559	$867	$2,282	$742	2.63	4.0	borderline
paid_search	2,338	$2,003,250	$857	$797	$221	0.93	>24	inefficient
social_ads	1,914	$1,779,919	$930	$398	$98	0.43	>24	inefficient

The CAC column and LTV/CAC ratio use the full observed window. Payback uses a separate CAC aligned to the mature subset's acquisition dates, preventing later spend from being compared with earlier cohort contribution.

Organic, referral, and partners clear the 3× policy and recover CAC within two acquisition-age months in the mature-customer curves. Paid search and social ads sit below the 1× line, meaning the average customer they buy is worth less than the cost to acquire them. The efficiency gap is not marginal. It is more than an order of magnitude between the best and worst channel. Payback uses the 19% to 22% of each channel's customers mature enough for the full 24-month horizon; social ads remains unrecovered at that boundary.

The efficiency gap is an order of magnitude

LTV/CAC ranked across channels against the 1x and 3x reference lines.



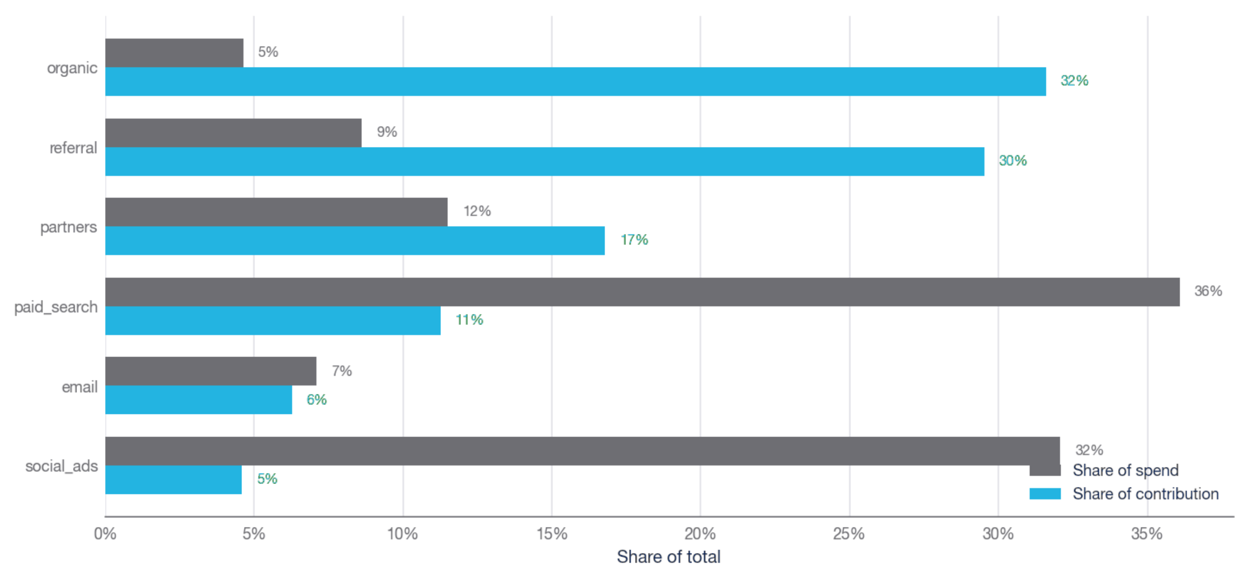
Source: unit_economics_channel_diagnostics.csv · synthetic data

Figure 9. LTV/CAC ranked across channels. Organic returns 20× its cost; social ads returns 0.43×. Email is the only borderline case.

The problem is not that the inefficient channels exist; it is their weight. Together they hold $3.78M of acquisition spend, 68% of the total, while returning under a dollar of contribution per dollar spent. The misallocation is clearest when each channel's share of spend is set against its share of contribution.

Spend and contribution are pointed in opposite directions

Each channel's share of acquisition spend against its share of contribution margin.



Source: unit_economics_channel_diagnostics.csv · synthetic data

Figure 10. Share of spend against share of contribution by channel. Organic takes 5% of spend and produces 32% of contribution; paid search and social ads take 68% of spend and produce 16% of contribution.

Email is borderline at 2.63× with a 4-month payback. It is a hold-and-fix case rather than a scale-or-cut one. The median LTV column tells a sharper version of the same story than the mean: social ads has a median LTV of $98 against a CAC of $930, so the typical customer it buys is deeply underwater even before averaging in the rare high-value account.

The mean and median both matter here. Organic and referral have high average LTV and materially higher median LTV than the paid channels, so the conclusion is not just being carried by a few large winners. Social ads is the opposite: average LTV is already below CAC, and median LTV is only $98. That makes it a poor scaling candidate even before attribution risk is considered.

Section 5.8 tests a bounded reduction in these two channels and reallocates the released acquisition budget under explicit response and capacity assumptions.

5.6 Margin concentrates in the lowest-rate segment and product

Profitability by segment carries a structural risk that a revenue view hides. The largest segment is also the least efficient.

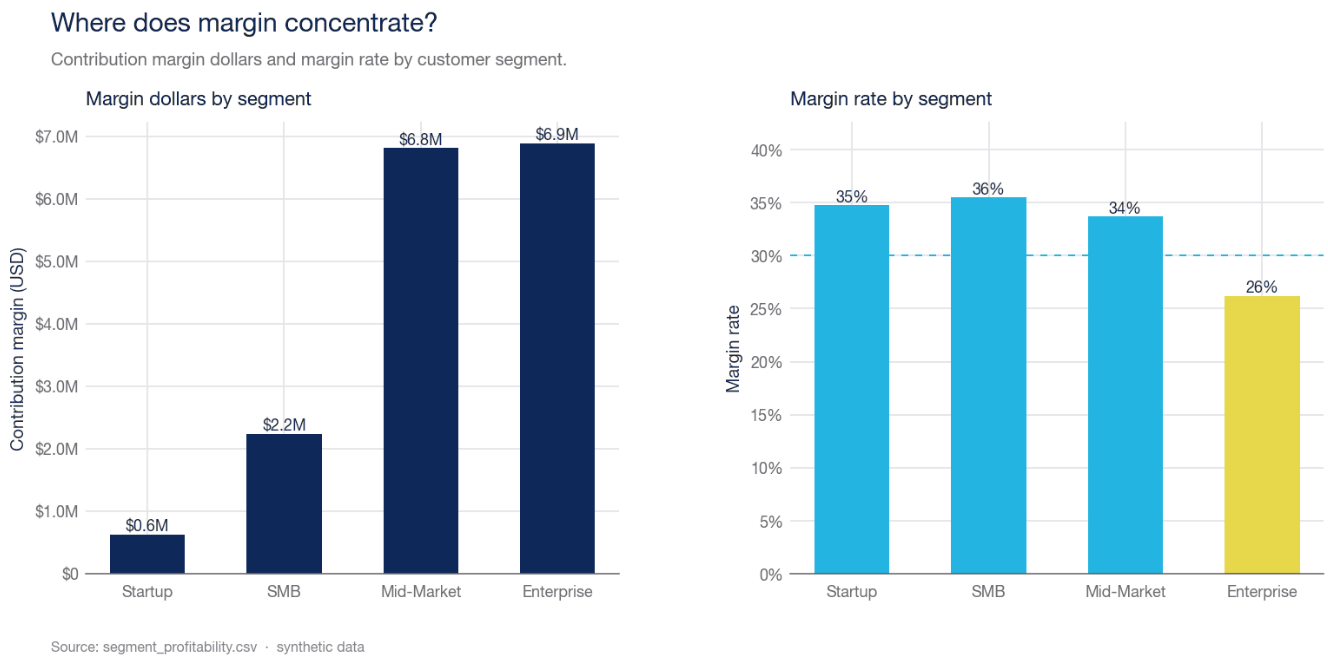


Figure 11. Contribution margin dollars and margin rate by segment. Enterprise carries the most margin dollars at the lowest rate, and it is the only segment below the 30% floor.

Segment	Customers	Revenue	Contribution margin	Margin rate	Revenue share
Enterprise	779	$26,282,659	$6,890,020	26.2%	48.1%
Mid-Market	2,415	$20,220,544	$6,814,179	33.7%	37.0%
SMB	3,184	$6,288,792	$2,232,644	35.5%	11.5%
Startup	2,622	$1,803,972	$627,187	34.8%	3.3%

Enterprise is the largest segment at 48% of revenue and $6.9M of margin dollars, yet it carries the lowest margin rate at 26.2%, below every smaller segment. Mid-Market, SMB, and Startup all run between 36% and 34%. Nearly half the revenue base sits in the least efficient segment, which is a concentration risk: pricing or cost-to-serve pressure in Enterprise hits the margin line harder than its rate alone suggests. Closing Enterprise to the 30% floor would be worth roughly $1.0M of contribution before any volume response, but that estimate should be treated as a sizing anchor, not an additive forecast.

Geography, by contrast, is not where the margin problem lives. Regional margin rates are banded within 1.0 points of each other, from 29.9% in LATAM to 30.3% in North America. Regions differ in size, not in efficiency.

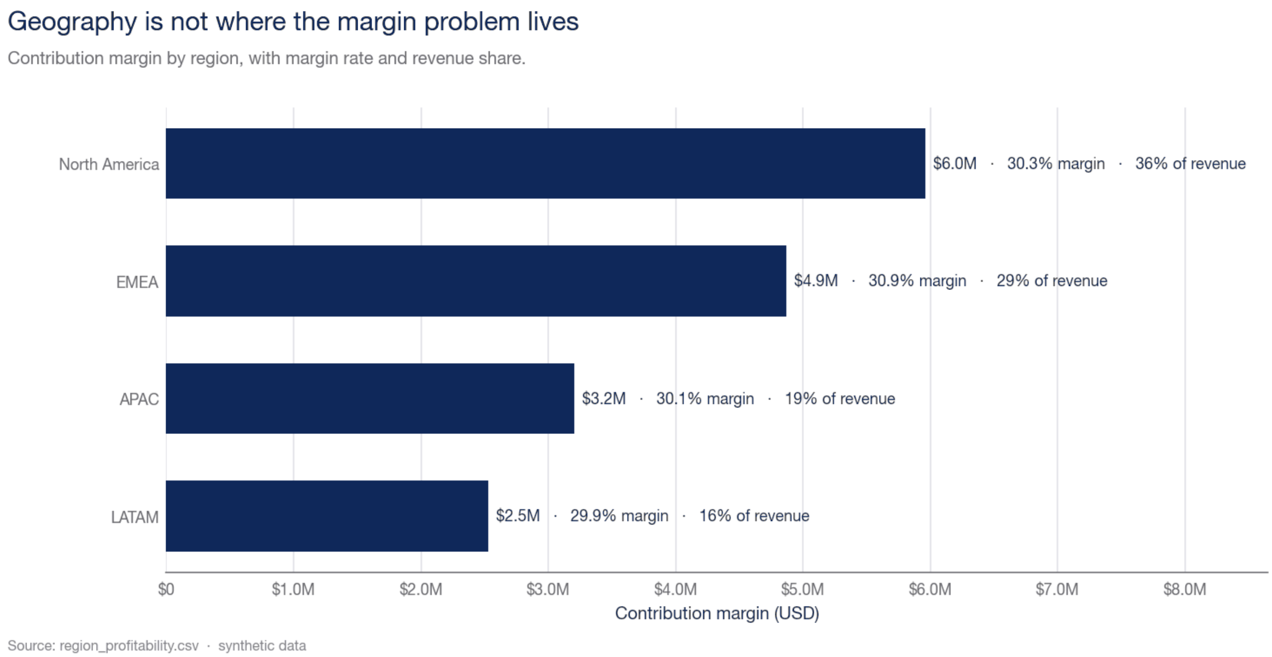


Figure 12. Contribution margin by region with margin rate and revenue share. North America leads on dollars, but every region sits within a point of the others on rate. Geography is a scale story, not a margin story.

The product view sharpens the diagnosis. Services run at just 17.7% margin on $13.6M of revenue (25% of the total), the deepest low-margin growth pocket. Premium is also below the 30% floor at 24.9%, while Core and Add-on, the two highest-rate lines, run at 41.5% and 47.6%. The spread between the best and worst product line is nearly 30 margin points.

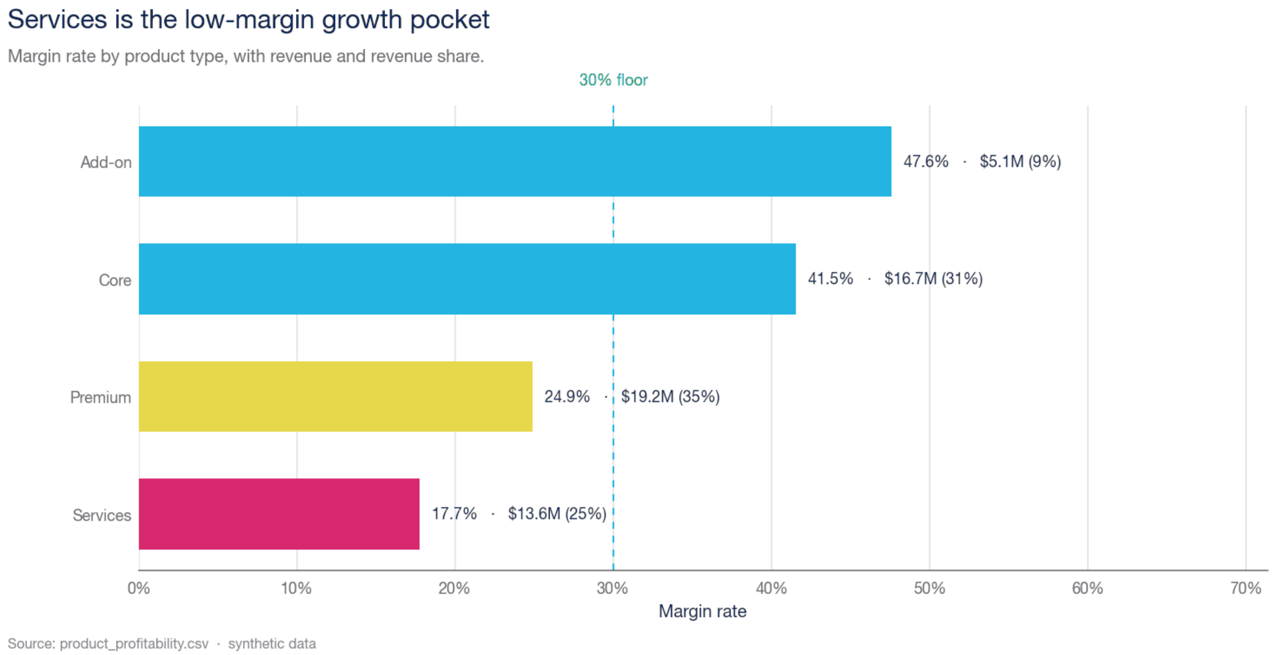


Figure 13. Margin rate by product type with revenue. Services is the deepest gap to the floor, Premium is a secondary gap, and Core plus Add-on are the products protecting the blend.

Taken together, the segment and product views say the margin problem is a mix and cost-to-serve problem, not a pricing-everywhere problem. The fix is targeted: re-price or re-scope the Services line and the Enterprise delivery model, rather than raising prices across a base that is already running at or above the floor everywhere else.

Do not add the segment and product gaps together. They are different cuts of the same underlying customers and transactions. Their value is diagnostic, not arithmetic: both cuts point to the same commercial workstream, which is Enterprise delivery economics and Services/Premium margin design. Within the product cut alone, closing Services and Premium to the 30% floor would address roughly $2.6M of margin gap before any behavioral response.

5.7 Revenue is concentrated in a thin band of customers

Revenue is far from evenly spread. The top 10% of customers account for 64% of total revenue, the top 20% for 81%, and the top 1% alone for 18%. The Gini coefficient of lifetime revenue is 0.77, well into the territory where a small group of accounts carries the business.

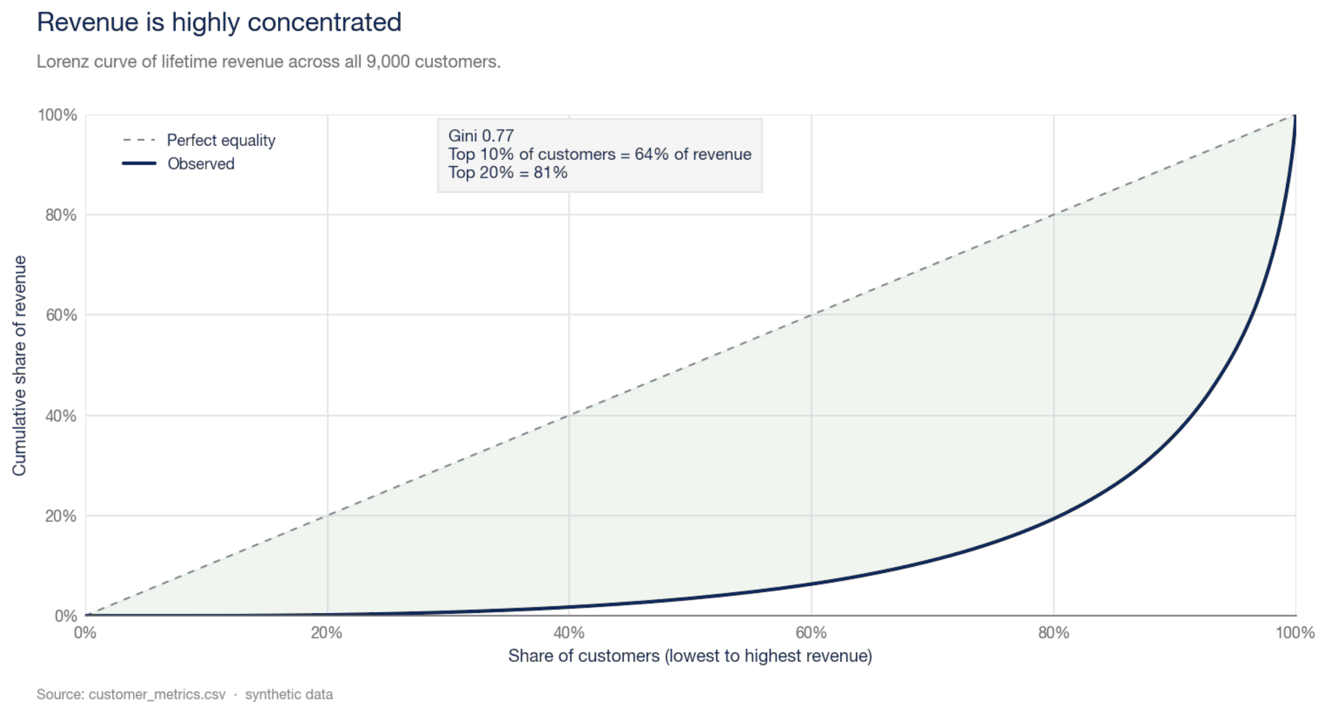


Figure 14. Lorenz curve of lifetime revenue across all 9,000 customers. The deep bow away from the equality line is the concentration: a fifth of customers hold four fifths of revenue.

The distribution behind the curve is a long right tail. Median observed-window revenue is $1,362 while the mean is $6,066, 4.5 times higher, because a small number of customers reach up to $266K each. The 843 customers who never transacted sit at the bottom of the same distribution and pull the median down further.

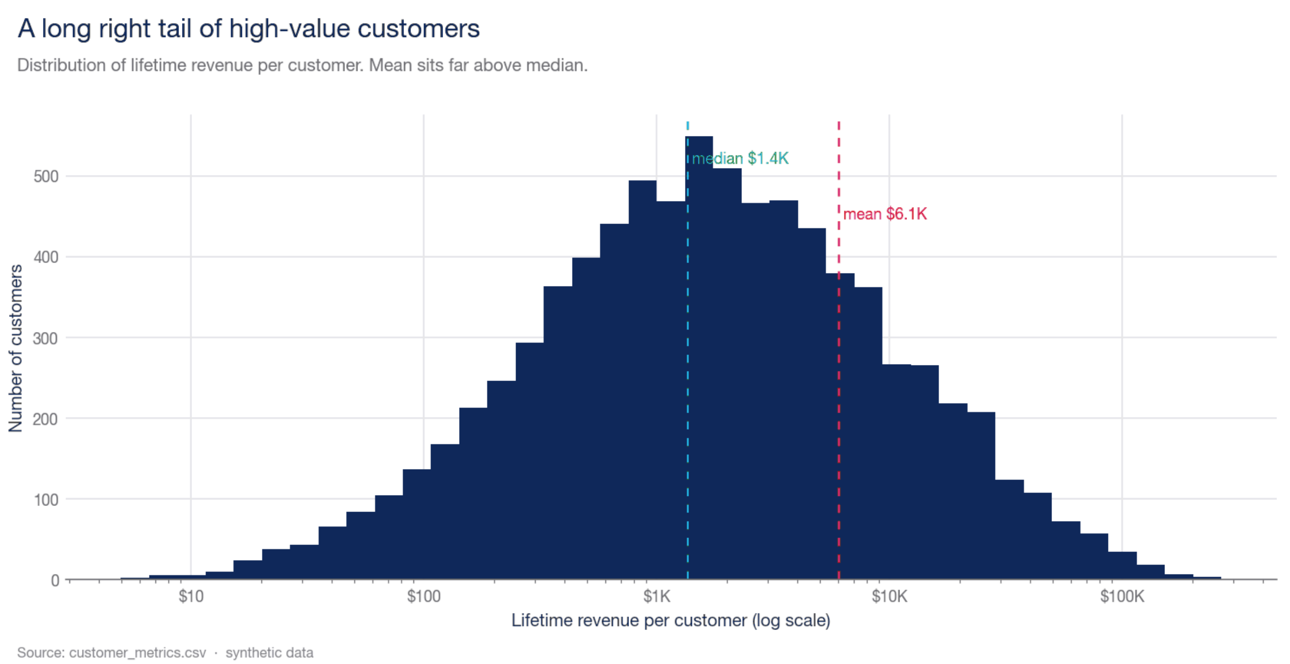


Figure 15. Distribution of observed-window revenue per customer on a log scale. The mean sits far to the right of the median, the signature of a heavy tail.

Concentration changes the risk profile because a small set of customers carries much of the observed revenue. Transaction activity span has a Pearson correlation of 0.42 with observed customer revenue, and transaction count has a correlation of 0.71. Both associations are partly mechanical: more observed transactions create both a longer possible span and more accumulated revenue.

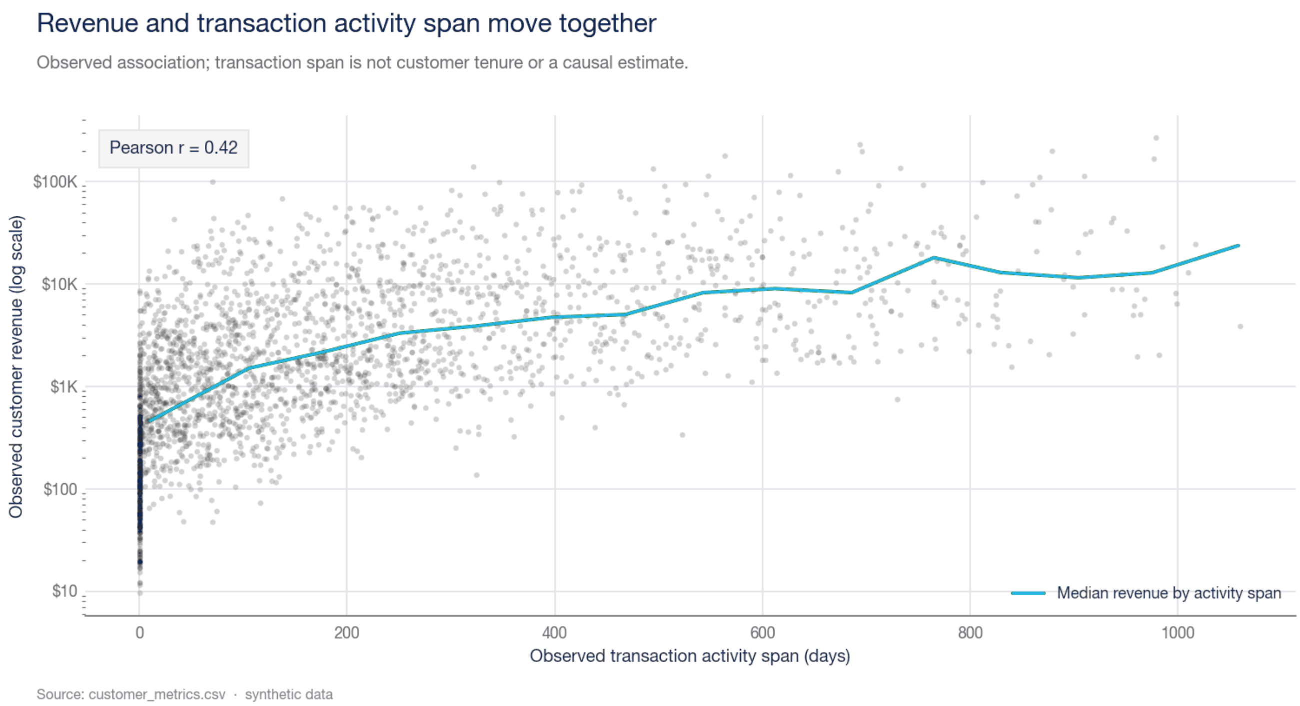


Figure 16. Observed revenue against first-to-last transaction span, with the median trend. This is an association, not customer tenure or a causal retention estimate.

The operating follow-up is to join high-value-customer concentration to channel and cohort evidence. The current aggregate views support that analysis but do not establish which channels cause longer or higher-value relationships.

The concentration read also protects the analysis from a common mistake: optimizing for the average customer when the business is carried by a small high-value tail. The right operating lens is not just whether a channel buys customers cheaply; it is whether later cohort evidence confirms durable contribution at the margin.

5.8 Reallocation models $7.9M uplift at the same acquisition budget

The reallocation scenario scales organic, referral, and partners within guardrails, cuts paid search and social ads by 35%, holds email, and reinvests the freed budget into the efficient channels up to a 100% scale-up cap per channel. Total budget is unchanged; the modeled uplift comes from the new allocation and the stated CAC/LTV response assumptions. The policy moves $1.3M of spend out of the two inefficient channels and assigns it only to channels that currently clear the LTV/CAC and payback gates. This does not model incremental operating costs outside acquisition spend.

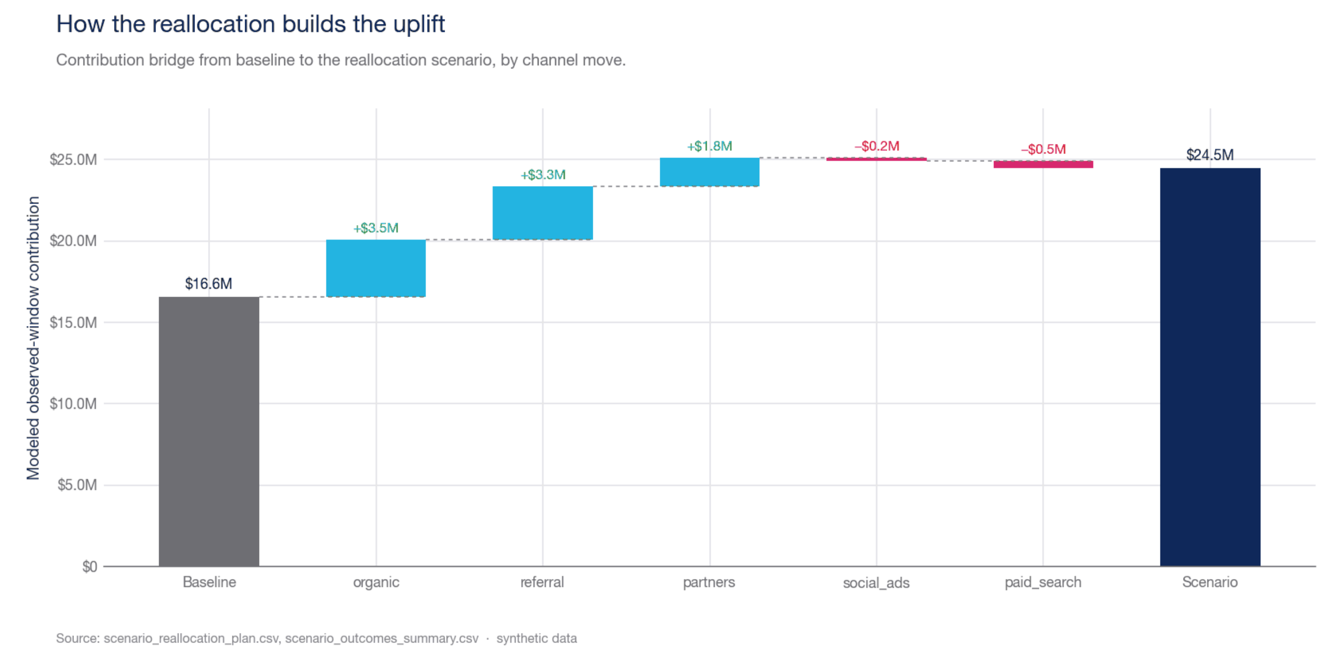


Figure 17. Contribution bridge from baseline to scenario by channel move. Scaling the three efficient channels adds the modeled uplift; trimming the two inefficient ones sacrifices some baseline contribution while releasing acquisition budget.

Channel	Status	Baseline spend	Scenario spend	Spend change	Contribution change
organic	efficient	$257,299	$514,599	+100%	+$3,519,556
referral	efficient	$478,649	$957,299	+100%	+$3,288,769
partners	efficient	$638,558	$1,226,718	+92%	+$1,750,014
email	borderline	$394,559	$394,559	+0%	$0
social_ads	inefficient	$1,779,919	$1,156,947	-35%	−$187,022
paid_search	inefficient	$2,003,250	$1,302,112	-35%	−$457,948

Modeled observed-window contribution rises from $16.6M to $24.5M in the base case, an uplift of $7.9M, or 48%. Uplift remains positive at the three specified stress points; this is sensitivity evidence, not a general robustness claim. CAC and LTV move against the policy in the worst case.

The economics of the bridge are concentrated. Scaling organic, referral, and partners contributes $8.6M of gross uplift, while cutting paid search and social ads removes $0.6M of contribution attached to weak traffic. That trade is still attractive because the freed spend earns materially more in the efficient channels than it loses in the inefficient ones.

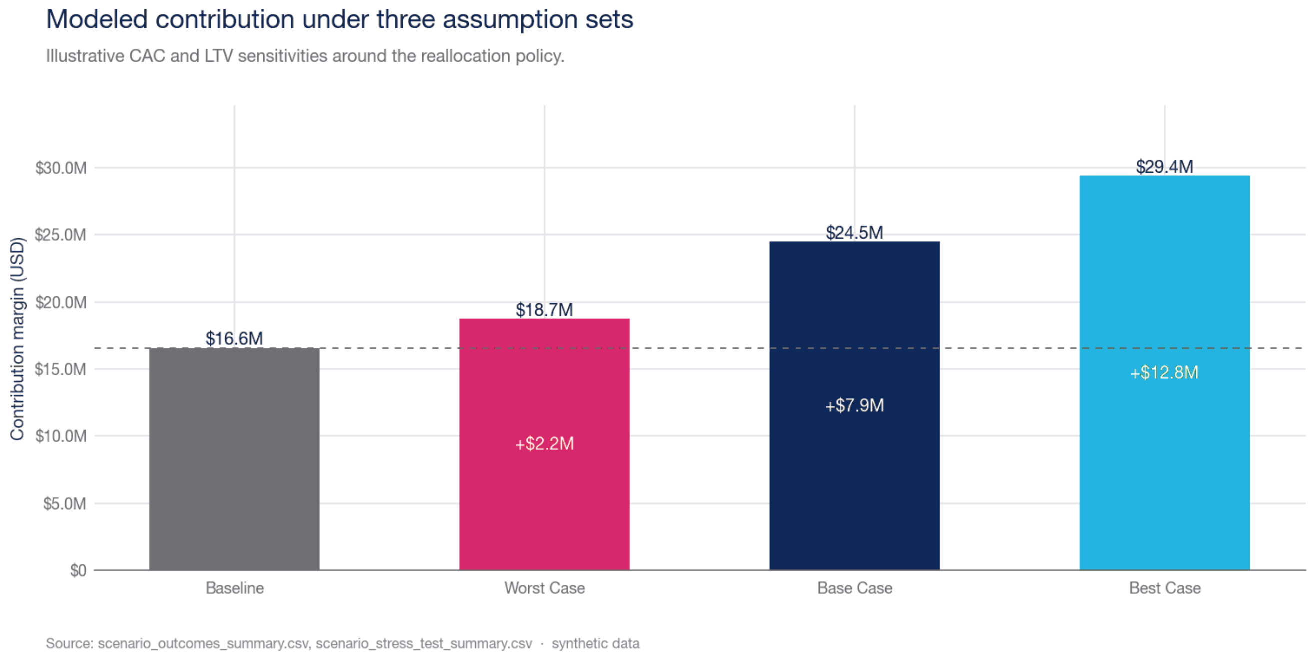


Figure 18. Contribution under the reallocation across stress cases. Every case clears the baseline; even the worst case, with CAC up 15% and LTV down 12% across the reallocated portfolio, adds $2.2M.

Scenario	CAC mult.	LTV mult.	Contribution	Uplift vs baseline
Best case	0.90x	1.08x	$29,372,878	+$12,808,848
Base case	1.00x	1.00x	$24,477,399	+$7,913,368
Worst case	1.15x	0.88x	$18,730,531	+$2,166,500

The worst case, with CAC inflated 15% and LTV cut 12% across every channel in the plan, still returns $2.2M above baseline. A separate same-generator sensitivity check repeats the workflow across 5 deterministic seeds. Uplift averages $8.0M with a standard deviation of $100K, a coefficient of variation of 1.3%, and it is positive in 5 of 5 seeds. The range runs from $7.8M to $8.1M.

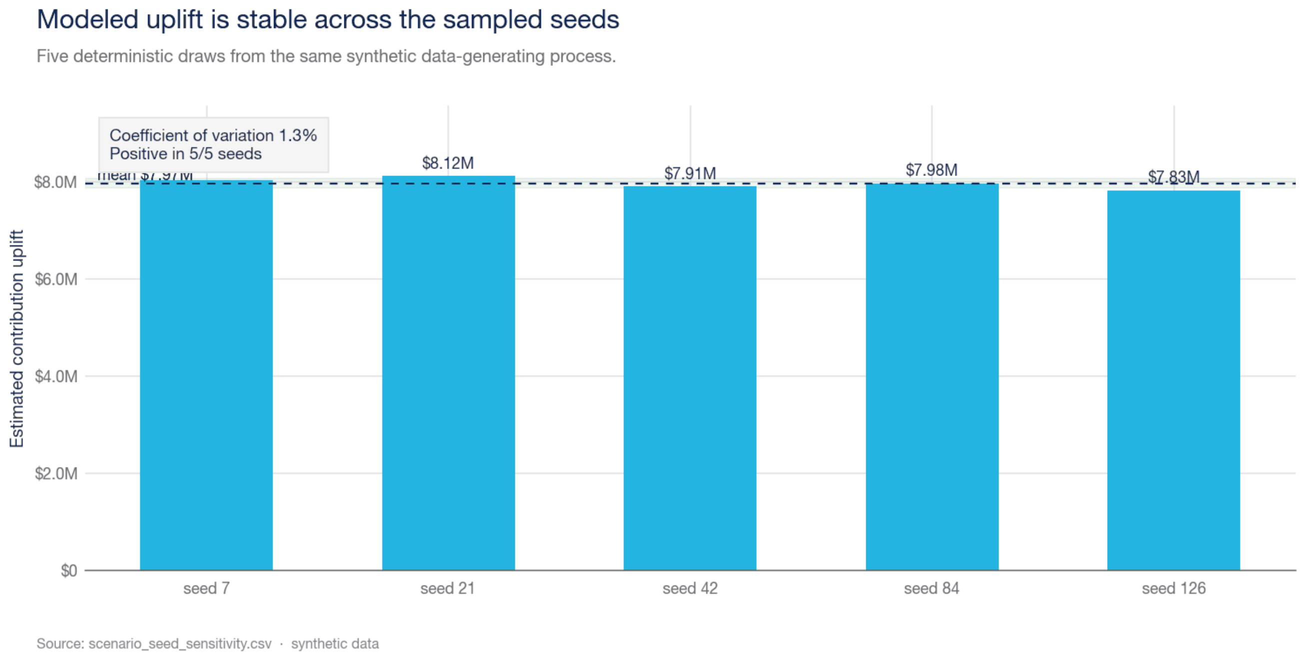


Figure 19. Estimated uplift across 5 deterministic draws from the same synthetic generator. This checks internal sensitivity, not external validity.

Seed	Baseline contrib.	Scenario contrib.	Uplift	Top scale-up	Top cut
7	$16,931,189	$24,965,691	+$8,034,503	partners	paid_search
21	$16,575,274	$24,695,403	+$8,120,129	partners	paid_search
42	$16,564,031	$24,477,399	+$7,913,368	partners	paid_search
84	$16,415,282	$24,390,470	+$7,975,188	partners	paid_search
126	$16,571,098	$24,398,607	+$7,827,509	partners	paid_search

The reallocation trades one concentration risk for another. Section 5.7 showed revenue concentrated in a thin band of customers; this policy concentrates spend in a thin band of channels. Organic, referral, and partners hold 25% of acquisition spend today; under the scenario they hold 49%, and their share of channel contribution rises from 78% to 88%. Partners alone would carry 22% of total spend post-reallocation. The plan is correct given what LTV/CAC and payback say today, but it is a bet on three channels holding their economics at roughly double their current scale, not a diversified portfolio move. Section 8 therefore treats capacity and channel-level guardrails as live monitoring items rather than a one-time approval, and it is the reason a fifth control, on channel concentration, is added there alongside the four performance guardrails.

Any real rollout would need staged stop-loss rules: pause a scaled channel if LTV/CAC falls below 3.0, if payback moves beyond 12 months, or if month-6 revenue retention for newly acquired cohorts deteriorates versus the case baseline.

6 Risks, limitations, and caveats

The findings are only as good as the assumptions behind them. This section states the limits plainly so the recommendations in Section 7 are read with the right confidence.

The data is synthetic

Every figure in this report comes from a synthetic dataset. It is internally consistent and passes all 15 validation checks, which makes it suitable for demonstrating method and for relative comparison between channels, segments, and cohorts. It is not a forecast of any real market, and the absolute dollar figures should not be read as predictions. The structure of the findings, the direction of the channel split, and the mechanics of the reallocation are the transferable part.

Observed LTV is window-limited

Lifetime value is measured over the available window, not over a true customer lifetime. Channels that acquire younger cohorts have had less time to accumulate LTV, which can understate their true value. The observed ratios therefore support a case comparison, not a claim about full-lifetime economics.

Payback uses a mature-customer subset

Only 19% to 22% of customers by channel are old enough to contribute a complete 24-month curve. Mature zero-transaction customers remain in the denominator. Social ads is right-censored at >24 months; this differs from insufficient maturity, which would leave the classification undefined. The recovery threshold uses spend from the acquisition-date range represented by those mature customers rather than full-window CAC.

Attribution does not identify incrementality

Period CAC retains one governed acquisition channel per customer. A separate position-based model allocates observed contribution across pre-signup touches and reconciles the total, but its weights do not identify counterfactual channel impact. Randomized holdouts provide incremental marketing evidence. The reallocation remains bounded because those experiments cover specific treatments rather than every channel-budget response.

Product and segment gaps are diagnostic, not additive

The Enterprise, Services, and Premium margin gaps are different cuts of the same economic base. They should not be summed into one opportunity figure without a customer-product bridge. Their value is to identify where margin work should start: delivery model, service scope, packaging, discounting, and cost-to-serve in the slices that repeatedly sit below the floor.

The decomposition depends on the mix dimension

The growth decomposition uses segment as the mix dimension. Defining mix by region, product, or channel would allocate the 6% mix term differently. Volume is 69% under the segment specification; the decomposition should be rerun before asserting the same ordering under another mix dimension.

Channel response assumptions and measured price response are separate

The spend-reallocation scenario still applies bounded CAC and LTV response assumptions. Its $7.9M central value and $2.2M tested downside are not forecast bounds. Product price elasticity is measured from randomized weekly assignments, but supports decisions only inside the observed 0.90–1.10 price index and synthetic design.

Cohort maturity

Long-horizon retention reads draw only on cohorts old enough to have reached that age. The month 24 figure rests on fewer cohorts than the month 6 figure, so the tail of the retention curve is noisier than its head. The month 3 to 6 decay, which carries the recommendation, sits in the best-observed part of the curve.

Execution capacity is assumed

The reallocation assumes efficient channels can absorb additional spend up to the stated cap while remaining inside the modeled CAC and LTV penalties. That is plausible based on the observed gap, but not guaranteed. Capacity would need to be evaluated during rollout with CAC, payback, conversion quality, and cohort-retention checks rather than assumed for the full quarter.

Recommendations and action priorities

Within this synthetic case, the most direct decision is to test a bounded reallocation away from the two channels below 1× observed LTV/CAC. Activation, retention, and low-margin product/segment pockets require further diagnosis before intervention. The actions below separate modeled value from unsized follow-up work.

Read across, not down: the matrix below is the executive decision lens, and the numbered detail that follows is the substantiation for each row.

Recommendation	Modeled case value	Evidence	Pilot timing	Reversibility
P1 · Reallocate spend	$7.9M (range $2.2M–$12.8M)	Positive in three stress cases and 5 same-process seed draws	Staged test	High — budget-neutral, reversible
P2 · Diagnose activation and retention	Not directly sized	Aggregate cohort pattern; driver not identified	1–2 cohorts to first read	Medium — onboarding and process change
P3 · Diagnose Services/Enterprise margin	~$2.6M to the floor (sizing anchor, not additive)	Gap identified; price-vs-cost split unknown	Cost-to-serve analysis first	Low-medium — pricing and contract change
P4 · Hold email, run experiment	Bounded — 7% of total spend	High confidence in the test design, low in the outcome	1–2 quarters	High — small, contained test
P5 · Run bounded product price tests	$1,999 predicted weekly contribution	Randomized price elasticity; support limited to 0.90–1.10	4–8 weeks	High — candidates remain inside tested range

1. **PRIORITY 1** **Pilot a staged reduction in paid search and social ads.**
   Both return under 0.93× cost and together hold 68% of acquisition spend. Cut each by 35% and redirect the freed $1.3M into organic, referral, and partners, holding each to its LTV/CAC and payback guardrails. This is the 48% contribution uplift modeled in Section 5.8, worth $7.9M in the base case and at least $2.2M in the tested downside case. These values are scenario outputs under assumed response curves, not forecast returns.

2. **PRIORITY 2** **Separate activation, retained activity, and revenue retention by acquisition slice.**
   Median revenue retention moves from 88% at month 3 to 68% at month 6, while only 52% of signups activate in month 0. Break these measures down by channel, segment, and product before selecting an onboarding or lifecycle intervention.

3. **PRIORITY 3** **Decompose the Services and Enterprise margin gap into price, discount, scope, and cost-to-serve.**
   Services at 17.7% and Enterprise at 26.2% are where mix dilutes the blended rate. Re-price or re-scope the lowest-margin product and delivery slices only after that bridge is available. Segment and product gaps overlap and must not be added together.

4. **PRIORITY 4** **Hold email flat and run a focused economics experiment.**
   Email is borderline at 2.63× with a 4-month payback. Do not scale or cut it on the current evidence. Run CAC and

payback experiments for one or two quarters and decide its fate on the result. Apply the same guardrails to the efficient channels as they scale, since the scenario assumes diminishing returns and the rollout should respect them.

The reallocation is the only recommendation directly sized by the scenario. The other actions are evidence plans; their value is not estimated in this report.

Sequencing

A real implementation would begin with a limited reallocation tranche and a paid-channel holdout. In parallel, instrument the activation/retention slices and build the price-versus-cost bridge. Scale, hold, or reverse only after marginal cohort evidence is available.

8 Decision controls and open questions

The recommendation is not to declare the efficient channels permanently efficient. It is to move budget in a controlled way, then let observed cohort and payback evidence decide whether the next move is scale, hold, or reverse. The control system below is the minimum needed to protect the upside in Section 5.8.

Controls for the reallocation

Control	Trigger	Decision response
LTV/CAC guardrail	Any scaled channel drops below 3.0×	Freeze incremental budget and inspect customer mix before further scale
Payback guardrail	Any scaled channel moves beyond 12 months	Reduce spend to the last clean level and review CAC inflation
Retention guardrail	Month-6 revenue retention deteriorates versus the current 68% baseline	Shift focus from acquisition scale to onboarding and activation quality
Margin guardrail	Blended contribution margin remains pinned at the 30% floor after reallocation	Escalate Services, Premium, and Enterprise cost-to-serve work
Attribution guardrail	Paid search or social ads show material assist value in a multi-touch read	Reclassify from cut to constrained hold, with spend tied to assisted contribution
Channel concentration guardrail	Any single scaled channel exceeds 22% of total spend, or the three-channel group shows synchronized softening in the same quarter	Cap further scale-up to that channel and hold the freed increment as unallocated budget rather than force it into the remaining two

Open questions that could change the decision

How much incremental capacity do organic, referral, and partners really have? The scenario caps scale-up at 100%, but observed channel performance should be monitored by marginal cohort, not only by blended historical channel averages. The key question is whether the next tranche of spend keeps the same buyer quality as the existing base.

Do paid search and social ads remain below 1× after multi-touch attribution? Their observed LTV/CAC is weak enough to justify cuts, but multi-touch or assisted-conversion evidence could support a smaller holdout budget. The current action should be a reduction with measurement, not an irreversible shutdown.

Which customers are responsible for the early retention drop? The cohort curve proves the decay pattern; it does not yet isolate the exact product, segment, channel, or use-case driver. Because revenue is concentrated, the follow-up should prioritize high-value customers with early usage decay rather than treating all churn equally.

What part of the Enterprise and Services margin gap is price versus cost-to-serve? The report identifies where the gap sits, not the operating root cause. The next analysis should split gross margin leakage into discounting, delivery effort, support load, fulfilment cost, and product scope before changing price cards or service packages.

90-day evidence plan

Weeks 1-2: move the first tranche, instrument channel-level CAC, payback, and new-cohort quality, and create a holdout read for the reduced paid channels. **Weeks 3-6:** compare marginal cohorts against current channel baselines, with a separate read for high-value and Services-heavy journeys. **Weeks 7-12:** complete, pause, or reverse the full reallocation based on guardrail performance, not spend delivery alone.

9

Appendix

A. Channel unit economics (full)

Channel	Customers	Spend	CAC	Avg LTV	Med LTV	LTV/CAC	Payback evidence (mo)	Status
organic	1,736	$257,299	$148	$3,014	$984	20.33	0.0	efficient
referral	1,668	$478,649	$287	$2,931	$1,117	10.21	1.0	efficient
partners	889	$638,558	$718	$3,127	$1,085	4.35	3.0	efficient
email	455	$394,559	$867	$2,282	$742	2.63	4.0	borderline
paid_search	2,338	$2,003,250	$857	$797	$221	0.93	>24	inefficient
social_ads	1,914	$1,779,919	$930	$398	$98	0.43	>24	inefficient

B. Segment and region profitability

Segment	Customers	Revenue	Contribution margin	Margin rate	Revenue share
Enterprise	779	$26,282,659	$6,890,020	26.2%	48.1%
Mid-Market	2,415	$20,220,544	$6,814,179	33.7%	37.0%
SMB	3,184	$6,288,792	$2,232,644	35.5%	11.5%
Startup	2,622	$1,803,972	$627,187	34.8%	3.3%

Region	Revenue	Contribution margin	Margin rate	Revenue share
North America	$19,715,177	$5,964,048	30.3%	36.1%
EMEA	$15,778,734	$4,868,825	30.9%	28.9%
APAC	$10,638,073	$3,203,088	30.1%	19.5%
LATAM	$8,463,982	$2,528,069	29.9%	15.5%

C. Product profitability

Product type	Revenue	Contribution margin	Margin rate	Revenue share
Add-on	$5,142,972	$2,446,225	47.6%	9.4%
Core	$16,729,012	$6,945,826	41.5%	30.6%
Premium	$19,150,210	$4,764,307	24.9%	35.1%
Services	$13,573,772	$2,407,672	17.7%	24.9%

D. Monthly revenue health (full series)

The 36-month series behind Figures 1, 2, 3, and 4.

Month	Revenue	Contribution margin	Margin rate	Active cust.	Transactions	Rev. growth
Jan 2023	$65,387	$23,197	35.5%	67	126	
Feb 2023	$160,455	$51,287	32.0%	125	233	+145.4%
Mar 2023	$233,272	$79,958	34.3%	193	331	+45.4%
Apr 2023	$355,715	$102,498	28.8%	265	447	+52.5%
May 2023	$417,188	$117,628	28.2%	329	543	+17.3%
Jun 2023	$524,416	$147,813	28.2%	396	715	+25.7%
Jul 2023	$522,944	$157,436	30.1%	431	709	-0.3%
Aug 2023	$523,311	$157,384	30.1%	476	819	+0.1%

Month	Revenue	Contribution margin	Margin rate	Active cust.	Transactions	Rev. growth
Sep 2023	$684,967	$206,665	30.2%	526	960	+30.9%
Oct 2023	$783,936	$227,396	29.0%	550	976	+14.4%
Nov 2023	$923,082	$271,152	29.4%	632	1,126	+17.7%
Dec 2023	$1,018,738	$305,513	30.0%	671	1,206	+10.4%
Jan 2024	$869,209	$264,527	30.4%	714	1,251	-14.7%
Feb 2024	$1,172,528	$349,835	29.8%	770	1,400	+34.9%
Mar 2024	$1,125,870	$353,665	31.4%	892	1,570	-4.0%
Apr 2024	$1,162,940	$354,161	30.5%	963	1,659	+3.3%
May 2024	$1,429,033	$425,027	29.7%	1,017	1,792	+22.9%
Jun 2024	$1,409,157	$440,333	31.2%	1,023	1,862	-1.4%
Jul 2024	$1,551,185	$476,109	30.7%	1,069	1,875	+10.1%
Aug 2024	$1,648,498	$505,093	30.6%	1,134	2,014	+6.3%
Sep 2024	$1,770,296	$538,881	30.4%	1,170	2,048	+7.4%
Oct 2024	$1,804,459	$557,671	30.9%	1,244	2,309	+1.9%
Nov 2024	$1,703,617	$539,287	31.7%	1,266	2,268	-5.6%
Dec 2024	$1,911,900	$599,972	31.4%	1,392	2,434	+12.2%
Jan 2025	$2,032,353	$606,116	29.8%	1,404	2,503	+6.3%
Feb 2025	$2,076,299	$626,984	30.2%	1,499	2,583	+2.2%
Mar 2025	$2,416,792	$719,595	29.8%	1,649	2,974	+16.4%
Apr 2025	$2,466,828	$732,700	29.7%	1,725	3,109	+2.1%
May 2025	$2,346,196	$706,402	30.1%	1,779	3,180	-4.9%
Jun 2025	$2,641,011	$781,949	29.6%	1,839	3,226	+12.6%
Jul 2025	$2,658,000	$817,483	30.8%	1,916	3,429	+0.6%
Aug 2025	$2,693,582	$802,697	29.8%	1,943	3,435	+1.3%
Sep 2025	$2,767,966	$841,698	30.4%	1,963	3,488	+2.8%
Oct 2025	$2,868,488	$898,570	31.3%	2,010	3,686	+3.6%
Nov 2025	$2,732,280	$833,462	30.5%	2,000	3,637	-4.7%
Dec 2025	$3,124,066	$943,889	30.2%	2,245	4,027	+14.3%

E. Cohort retention curve (months 0 to 24)

Median signup activity, retained-from-month-0 activity, and revenue retention across cohorts at each observed age.

Months since signup	Signup activity	Retained from M0	Revenue retention	Cohorts observed
0	52.2%	100.0%	100.0%	36
1	50.3%	52.4%	91.6%	35
2	45.4%	47.8%	81.4%	34
3	43.3%	45.5%	88.4%	33
4	38.7%	39.0%	75.5%	32
5	36.8%	37.3%	67.6%	31
6	34.5%	36.0%	68.3%	30

Months since signup	Signup activity	Retained from M0	Revenue retention	Cohorts observed
7	30.7%	32.0%	56.0%	29
8	29.5%	30.4%	63.8%	28
9	27.3%	28.9%	55.3%	27
10	25.0%	26.8%	51.1%	26
11	22.5%	23.9%	47.6%	25
12	22.1%	24.1%	49.6%	24
13	21.1%	23.1%	42.7%	23
14	19.1%	19.4%	44.5%	22
15	17.5%	17.0%	37.1%	21
16	16.8%	16.8%	34.9%	20
17	15.7%	15.6%	34.5%	19
18	15.5%	17.0%	33.9%	18
19	14.3%	15.6%	34.9%	17
20	13.9%	12.6%	26.2%	16
21	12.7%	12.8%	29.0%	15
22	11.6%	11.7%	22.4%	14
23	11.8%	11.8%	23.1%	13
24	10.3%	9.4%	21.8%	12

F. Data validation gate

All 15 checks run on the raw tables before any analysis. A failure blocks the pipeline.

Check	Status	Detail
table_row_count_non_zero	PASS	customers=9,000, transactions=69,950, marketing_spend=6,576, marketing_touchpoints=22,465,
schema_columns_match	PASS	customers=['customer_id', 'signup_date', 'segment', 'region', 'acquisition_channel']; tran
null_values_raw_tables	PASS	total_raw_nulls=0
contract_types_and_finite_values	PASS	invalid_dates=0, invalid_or_nonfinite_numeric_values=0
grain_key_uniqueness	PASS	duplicate customer_id=0, duplicate transaction_id=0, duplicate date+channel=0, duplicate t
categorical_domain_values	PASS	unexpected_values=none
transaction_customer_referential_integrity	PASS	orphan_transaction_rows=0, orphan_touchpoint_rows=0, orphan_experiment_rows=0
transaction_date_not_before_signup	PASS	rows_with_transaction_before_signup=0
marketing_journey_integrity	PASS	post_signup_touchpoints=0, customers_without_exactly_one_conversion_touch=0, conversion_to
value_range_sanity	PASS	non_positive_revenue=0, negative_cost=0, negative_spend=0
causal_design_input_integrity	PASS	negative_experiment_values=0, experiments_missing_arm=0, negative_pricing_values=0, invali
negative_margin_transaction_review	PASS	rows_with_cost_above_revenue=258, share=0.37%, review_threshold=1.00%
channel_domain_alignment	PASS	customer_channels=['email', 'organic', 'paid_search', 'partners', 'referral', 'social_ads'

Check	Status	Detail
marketing_date_channel_coverage	PASS	expected_pairs=6,576, observed_pairs=6,576, missing_pairs=0
date_coverage_observed	PASS	customers=2023-01-01..2025-12-31, transactions=2023-01-03..2025-12-31, marketing=2023-01-0

G. Data dictionary and reproducibility

Field	Definition
contribution_margin	Revenue minus direct delivery cost
average_LTV	Mean cumulative contribution margin per acquired customer, including zero-transaction customers
median_LTV	Median cumulative contribution margin per acquired customer
CAC	Period channel spend divided by customers acquired in the channel
payback_CAC	Spend inside the mature customers' acquisition-date window divided by those mature customers; used only for the empirical payback curve
LTV_to_CAC	average_LTV divided by CAC; efficient at ≥ 3.0
approximate_payback_period	First acquisition-age month when cumulative contribution per mature customer recovers payback_CAC; unrecovered cases are right-censored at >24 months
median_month_0_activation_rate	Median share of signups active in their signup month
median_signup_activity_rate	Median share of the original signup cohort active at age m
median_retained_from_month_0_rate	Median share of month-0 active customers also active at age m
median_revenue_retention	Median cohort revenue at month m relative to month 0
incremental_contribution_per_treated_customer	CUPED-adjusted treatment-minus-control contribution from a randomized holdout
price_elasticity	Log demand response to log price inside the randomized 0.90–1.10 price range
attributed_contribution	Observed contribution allocated by position-based journey weights; descriptive, not causal
Gini coefficient	Concentration of observed-window revenue across customers; 0 is even, 1 is fully concentrated